



1. Collection Datatypes

- **Lists & Dictionaries:** Understanding core collection types and their usage.

2. Loops & Comprehensions

- **Loops:** Using `for` and `while` loops effectively.
- **Comprehensions:** Introduction to list and dictionary comprehensions for concise coding.

3. Advanced Comprehensions

- **Conditional Comprehensions:** Add conditions to comprehensions for filtered output.
- **Zip, Enumerate & Nested Comprehensions:** Combining and iterating through multiple collections.

4. File Handling & NumPy



- **Files Read/Write:** Reading from and writing to files.
- **NumPy:** Working with numerical data and arrays.

5. Pandas: Read, Select & Filter

- **DataFrames:** Reading data into Pandas, selecting columns, and filtering rows.

6. Data Cleaning

- **Handling Missing Data:** Identifying and addressing missing values.
- **Removing Duplicates:** Clean your dataset of redundant records.
- **Transforming Data:** Basic transformation techniques.

7. Data Cleaning Continued

- **Replacing Values:** Update specific values in your dataset.
- **Renaming Axis Labels:** Renaming columns or index for better understanding.
- **Dealing with Outliers:** Identifying and managing outliers.

8. Data Cleaning: Transformations

- **Numerical to Categorical:** Converting numerical data into categorical types.
- **Categorical to Numerical:** Encoding categories into numerical values.

9. Data Cleaning: Strings & Regex

- **String Manipulation:** Handling textual data.
- **Regular Expressions (Regex):** Advanced string processing.

10. Data Aggregation & Exploration

- **Aggregating Data:** Summarize and explore datasets with statistical operations.

11. Data Visualization

- **Matplotlib:** Visualizing data using Python's popular library.
 - **Bar, Pie, Line, Histogram, Scatter:** Different types of charts.
 - **Subplots:** Display multiple visualizations in a single figure.

12. Case Study

- **Application:** Apply all learned concepts to analyze a real-world dataset.
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